Shivani Toorpu

github.com/Shivani-Toorpu linkedin.com/in/shivani-toorpu +91 63005 47803 toorpushivani@gmail.com

Summary

Computer Science graduate skilled in Python, Node.js, ASP.NET Core, DSA, and core CS fundamentals. Seeking entry-level Software Engineer/Developer roles to apply clean coding practices, leverage AI tools, and deliver scalable solutions.

Education

Bachelor of Technology in Computer Science and Engineering (AI & ML) *May 2026*
Vardhaman College of Engineering, Shamshabad, Hyderabad, Telangana. CGPA: 8.69

Work Experience

Full-Stack Developer (Intern) | Link *Jan 2025 – Mar 2025*
Innomatics Research Labs Online

- Architected responsive React.js frontends and scalable Node.js/Express.js backends integrated with MongoDB Atlas.
- Integrated RESTful APIs with JWT authentication and bcrypt hashing to handle complex CRUD operations and data security.
- Performed debugging, performance optimization, and version control using Git/GitHub before deploying production-ready builds to Vercel.

Projects

QuickRoute – URL Shortener Service | C#, ASP.NET Core, PostgreSQL | Link *Jan 2026 - Mar 2026*
A high-throughput URL shortening service with in-memory caching and analytics, run locally.

- Engineered a URL shortener using Base62 encoding on auto-incremented IDs, sustaining 100+ req/s with zero failures.
- Designed a custom LRU cache from scratch using HashMap + Doubly LinkedList achieving 99% cache hit rate, serving most redirects entirely from memory without touching the database.
- Instrumented with structured logging and a live /stats endpoint exposing real-time cache hit rate for performance observability.

Sorting Algorithm Analyzer | C++, STL, React | Link *Aug 2025 - Oct 2025*
An interactive sorting algorithm visualizer with a C++ benchmarking engine and a React-based animated UI.

- Implemented 5 sorting algorithms across modular files, measuring comparisons, swaps, and execution time to analyze time complexity tradeoffs.
- Engineered a React UI with step-by-step animation and play/pause/step controls to expose algorithm behavior on custom input.
- Built a benchmarking table running all 5 algorithms on identical input, surfacing performance winners per algorithm.

Healthcare Booking System | JavaScript, React.js, Node.js | Link *Feb 2025 - Jun 2025*
Full-stack healthcare appointment platform with role-based access and dashboards for patients, doctors, and admins.

- Built secure backend RESTful APIs with Node.js and Express.js using JWT authentication and bcrypt password encryption.
- Developed a responsive frontend with React Router protected navigation and Axios interceptors for automated token injection.
- Designed MongoDB Atlas data models to support doctor approval pipelines, slot availability tracking, and specialty-based appointment booking.

Technical Skills

Languages: Python, C#, Java, C++, JavaScript, TypeScript, SQL
Frameworks & Libraries: ASP.NET Core, React.js, Node.js, Express.js, Next.js, Tailwind CSS, REST APIs, JWT
Databases & Cloud: PostgreSQL, MySQL, MongoDB, Redis, AWS
Tools: Git, GitHub, Postman, Linux, Vercel, Swagger
Engineering Practices: Code Reviews, Debugging, Version Control, API Testing, Documentation, AI-assisted Development
Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Software Engineering, Computer Networks, Cloud Computing, System Design

Certifications

AWS Academy Graduate – Cloud Foundations, AWS Training and Certification
Oracle Cloud Infrastructure AI Foundations Associate, Oracle
Problem-Solving (Intermediate), **SQL (Basic)**, HackerRank

Coding Profiles

LeetCode (Shivani_Toorpu): 400+ problems solved, 1700+ contest rating
HackerRank (Toorpu_Shivani): 4★ in Problem-Solving
CodeChef (shivani_toorpu): 150+ problems solved, 93% score in C++ Skill Test

Achievements

IEEE ICICNCT 2025: Published research on Fake News Detection using LLMs in an IEEE Xplore indexed conference.
Amazon ML Challenge 2024: Led a team of 4 to extract entity values from product images, ranking 442/75,000+ teams.
Summer Analytics 2024: Placed in the top 25% in a data science and machine learning program by IIT Guwahati.